FedEx

Logistics Operations Analysis

Exploratory Data Analysis (EDA)

SUBMITTED BY
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PROJECT TYPE
Individual Project

DOMAIN
Logistics & Supply Chain Analytics

TOOLS USED

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Plotly





Analyze. | Insight. | Optimize. | Deliver.

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1. Executive Summary

The **FedEx Logistics Operations Analysis** project is an Exploratory Data Analysis (EDA) initiative designed to evaluate shipment operations, transportation performance, customer distribution, and overall logistics efficiency using historical shipment data from the Supply Chain Management System (SCMS). The primary objective of this project is to transform raw logistics data into meaningful business intelligence that supports informed decision-making, enhances operational efficiency, and strengthens supply chain performance.

FedEx is a globally recognized leader in logistics and transportation, providing reliable domestic and international shipping solutions across diverse industries. With the continuous growth of global trade and e-commerce, logistics organizations face increasing challenges such as rising transportation costs, shipment delays, inefficient routing, uneven regional demand, and growing customer expectations. Addressing these challenges requires data-driven decision-making to identify operational trends, optimize logistics resources, and improve service quality.

The analysis is based on a dataset containing **10,324 shipment records** and **33 business attributes**, including shipment destinations, transportation modes, freight costs, shipment weights, product categories, vendors, manufacturing sites, insurance values, delivery schedules, and related operational variables. The project was developed using **Python** in **Google Colab**, leveraging powerful data analysis and visualization libraries such as **Pandas**, **NumPy**, **Matplotlib**, **Seaborn**, and **Plotly**. A structured analytical workflow was followed, consisting of data collection, data understanding, preprocessing, missing value treatment, statistical analysis, exploratory data analysis, visualization, and business interpretation.

Comprehensive exploratory analysis was performed using a wide range of visualization techniques, including bar charts, histograms, box plots, scatter plots, violin plots, bubble charts, donut charts, correlation heatmaps, pair plots, and lollipop charts. These visualizations enabled the identification of hidden patterns, relationships, and trends within the logistics data, providing valuable insights into shipment distribution, freight cost behavior, transportation mode utilization, vendor performance, manufacturing site contributions, and regional shipment activity.

The analysis revealed several significant business insights. Shipment weight demonstrates a strong positive relationship with freight cost, indicating that heavier shipments generally incur higher transportation expenses. Air transportation emerged as the most frequently used shipment mode, reflecting the company's emphasis on faster delivery services while also contributing to increased logistics costs. Countries such as **South Africa**, **Nigeria**, and **Côte d'Ivoire** account for a substantial share of total shipments, highlighting their strategic importance within FedEx's global distribution network. Furthermore, a strong correlation was observed between **Line Item Value** and **Insurance Cost**, emphasizing the need for higher insurance coverage and effective risk management for high-value shipments.

Based on these findings, several business recommendations have been proposed, including optimizing high-volume shipping routes, improving freight cost management through shipment consolidation, balancing transportation modes according to business requirements, strengthening operations in high-demand regions, expanding services into emerging markets, and implementing predictive analytics for demand forecasting and transportation cost

estimation. The adoption of interactive business dashboards and data-driven decision-making frameworks can further improve operational visibility, resource utilization, and supply chain efficiency.

In conclusion, this project demonstrates the significant value of Exploratory Data Analysis in transforming complex logistics data into actionable business insights. The findings provide a strong foundation for enhancing operational efficiency, reducing transportation costs, improving customer satisfaction, and supporting strategic planning. Additionally, the project establishes a pathway for future integration of machine learning, predictive analytics, and real-time logistics monitoring, enabling FedEx to further strengthen its competitive advantage in the rapidly evolving global logistics and supply chain industry.

2. Introduction

In today's highly interconnected global economy, logistics and supply chain management play a vital role in ensuring the efficient movement of goods from manufacturers to customers. Effective logistics operations are essential for maintaining product availability, minimizing transportation costs, meeting delivery schedules, and achieving high levels of customer satisfaction. As supply chain networks continue to expand in scale and complexity, organizations increasingly rely on data analytics to monitor operational performance, identify inefficiencies, and make informed business decisions.

FedEx is one of the world's leading logistics and transportation companies, providing reliable domestic and international shipping, freight, and supply chain solutions across numerous countries. Managing millions of shipments through an extensive global network requires continuous monitoring of transportation performance, shipment tracking, delivery timelines, freight costs, and customer demand. Analyzing operational data enables the organization to optimize logistics processes, improve resource utilization, reduce operational expenses, and enhance service quality.

The **FedEx Logistics Operations Analysis** project applies **Exploratory Data Analysis (EDA)** techniques to examine historical shipment data obtained from the Supply Chain Management System (SCMS). The study focuses on key logistics variables, including shipment destinations, transportation modes, freight costs, shipment weights, vendors, manufacturing sites, product categories, delivery schedules, and insurance values. The analysis was performed using **Python** in **Google Colab**, supported by powerful data analysis and visualization libraries such as **Pandas**, **NumPy**, **Matplotlib**, **Seaborn**, and **Plotly**.

The primary objective of this project is to explore the dataset, identify underlying trends and patterns, detect operational inefficiencies, assess data quality, and generate meaningful business insights through statistical analysis and visualization. The findings are intended to support logistics optimization, improve transportation planning, reduce operational costs, strengthen supply chain efficiency, and facilitate data-driven strategic decision-making.

Overall, this project demonstrates the practical application of Exploratory Data Analysis in transforming large volumes of logistics data into actionable business intelligence. The insights generated provide valuable support for improving operational performance, enhancing customer service, optimizing resource allocation, and enabling sustainable growth within modern logistics and supply chain management.

3. Company Overview

FedEx Corporation is one of the world's leading logistics, transportation, and supply chain management companies, headquartered in **Memphis, Tennessee, USA**. Established in **1971** by **Frederick W. Smith**, the company revolutionized the express delivery industry by introducing an innovative overnight shipping network that transformed global package transportation. Over the years, FedEx has expanded its operations into a comprehensive logistics enterprise, serving customers across more than **220 countries and territories**.

The company offers a wide range of services, including express parcel delivery, ground transportation, freight forwarding, e-commerce logistics, supply chain solutions, customs brokerage, and last-mile delivery. Through its extensive global network of aircraft, vehicles, distribution hubs, and advanced tracking systems, FedEx manages millions of shipments every day while ensuring reliable, secure, and time-sensitive delivery services.

FedEx serves organizations of all sizes across diverse industries, including healthcare, manufacturing, retail, automotive, technology, aerospace, and consumer goods. Its integrated logistics solutions enable businesses to optimize inventory management, streamline supply chain operations, reduce transportation costs, and improve overall operational efficiency. The company's commitment to innovation is reflected in its adoption of digital technologies, automation, artificial intelligence, real-time shipment tracking, and data-driven decision-making to enhance service quality and operational performance.

With a strong focus on customer satisfaction, operational excellence, and continuous improvement, FedEx continues to invest in technology, infrastructure, and sustainable logistics practices. By leveraging advanced analytics and intelligent transportation systems, the company enhances delivery reliability, minimizes operational delays, optimizes resource utilization, and strengthens its competitive position in the rapidly evolving global logistics and supply chain industry.

4. Industry Overview

The **logistics and supply chain industry** plays a critical role in supporting global trade by ensuring the efficient movement of goods from manufacturers to customers. Rapid growth in e-commerce, globalization, and increasing customer expectations for faster deliveries have significantly accelerated the demand for advanced logistics services. Modern logistics companies rely on integrated transportation, warehousing, inventory management, and real-time tracking systems to optimize operations and reduce costs. Emerging technologies such as data analytics, artificial intelligence (AI), automation, the Internet of Things (IoT), and predictive analytics have transformed the industry by improving route optimization, demand forecasting, supply chain visibility, and operational efficiency. Despite these advancements, the industry continues to face challenges including rising transportation costs, fuel price fluctuations, delivery delays, capacity constraints, and supply chain disruptions. As a global leader, **FedEx Corporation** leverages digital innovation and data-driven decision-making to enhance operational performance, improve customer satisfaction, and maintain a competitive advantage in the rapidly evolving logistics sector.

5. Problem Statement

FedEx operates one of the world's largest logistics and transportation networks, handling thousands of shipments across multiple countries, transportation modes, vendors, manufacturing sites, and customer locations. As shipment volumes continue to grow, managing logistics operations efficiently while maintaining high service quality has become increasingly challenging. Operational issues such as rising freight costs, delivery delays, uneven regional performance, inefficient utilization of transportation modes, and supply chain bottlenecks can negatively impact operational efficiency, customer satisfaction, and overall business performance.

To address these challenges, it is essential to analyze historical shipment data to identify operational trends, understand the factors influencing transportation costs, evaluate shipment performance, and examine regional distribution patterns. By applying Exploratory Data Analysis (EDA), this project aims to uncover hidden patterns, detect data quality issues, identify operational inefficiencies, and generate actionable business insights. The findings will support data-driven decision-making, improve logistics planning, optimize transportation strategies, reduce operational costs, and enhance the overall efficiency and reliability of the supply chain.

6. Project Objectives

The primary objective of this project is to analyze FedEx shipment data using **Exploratory Data Analysis (EDA)** to gain meaningful insights into the company's logistics and supply chain operations. The study aims to transform raw shipment data into actionable business intelligence that supports operational improvement and strategic decision-making.

The specific objectives of the project are:

- To analyze shipment distribution across different countries, regions, and destinations.
- To evaluate the performance of logistics operations and identify key shipment trends.
- To compare transportation modes and examine their impact on delivery efficiency and freight costs.
- To identify the major factors influencing freight charges, shipment weight, and insurance costs.
- To assess vendor and manufacturing site performance based on shipment activities.
- To detect missing values, inconsistencies, and other data quality issues within the dataset.
- To explore relationships and correlations among business variables using statistical analysis and data visualization.
- To generate actionable business insights that support logistics optimization, cost reduction, and data-driven decision-making.

7. Project Scope

This project focuses on conducting an **Exploratory Data Analysis (EDA)** of the **FedEx Supply Chain Logistics Dataset** to understand shipment operations, transportation performance, and overall supply chain efficiency. The scope includes data collection, preprocessing, cleaning, statistical analysis, and visualization using **Python** in **Google Colab** with libraries such as **Pandas, NumPy, Matplotlib, Seaborn, and Plotly**. The analysis examines key business variables, including shipment destinations, transportation modes, vendors, manufacturing sites, freight costs, shipment weights, line item values, insurance costs, and customer distribution. It also involves identifying missing values, data quality issues, outliers, and relationships among variables through descriptive statistics and visual analytics. Based on the findings, the project provides actionable business insights and recommendations to optimize transportation strategies, improve operational efficiency, reduce logistics costs, enhance delivery performance, and support data-driven decision-making. The project establishes a foundation for future applications of predictive analytics and machine learning in logistics and supply chain management.

8. Dataset Description

This project utilizes the **SCMS (Supply Chain Management System) Delivery History Dataset**, a publicly available logistics dataset containing historical shipment records from the **USAID Global Health Supply Chain Program**. The dataset captures various aspects of logistics and transportation operations, including shipment details, destination countries, vendors, manufacturing sites, transportation modes, product categories, freight costs, shipment weights, delivery schedules, insurance values, and line item information.

The dataset contains **10,324 shipment records** and **33 attributes**, providing a comprehensive representation of supply chain activities. It includes both numerical and categorical variables, making it suitable for statistical analysis, data visualization, and business intelligence generation. The data enables detailed examination of shipment distribution, transportation performance, freight cost behavior, customer demand, and operational efficiency across different regions and logistics processes.

Dataset Summary

Attribute	Details
Dataset Name	SCMS Delivery History Dataset
Domain	Logistics & Supply Chain Analytics
Total Records	10,324
Total Features	33
Numerical Features	7
Categorical Features	26
Duplicate Records	0
Missing Values	2,383
Memory Usage	16.04 MB

Key Variables

Shipment Information

- Shipment Mode
- Scheduled Delivery Date
- Delivered Date
- Delivery Recorded Date

Location Details

- Country
- Manufacturing Site

Product Information

- Product Group
- Vendor
- Item Description
- Brand
- Dosage Form

Cost-Related Variables

- Freight Cost (USD)
- Line Item Value
- Pack Price
- Unit Price
- Insurance Cost

Quantity and Weight Variables

- Line Item Quantity
- Unit of Measure
- Weight (Kilograms)

Overall, this dataset provides a detailed view of logistics and supply chain operations, serving as the foundation for identifying shipment trends, transportation patterns, cost drivers, operational bottlenecks, and opportunities for process optimization through Exploratory Data Analysis (EDA). The insights generated from this analysis support data-driven decision-making and contribute to improving logistics performance and overall supply chain efficiency.

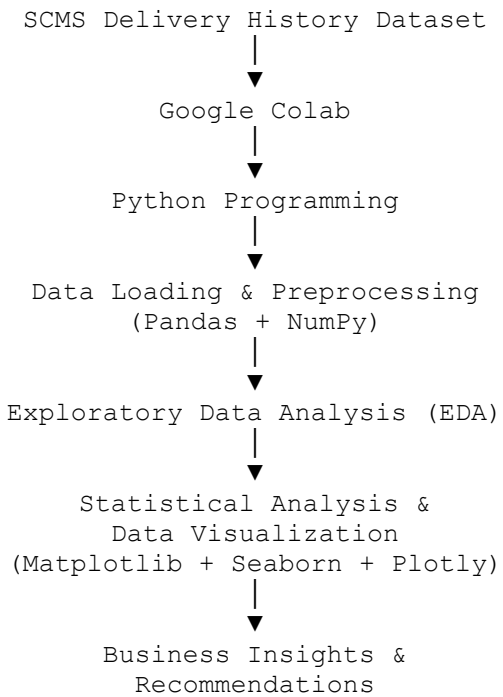
9. Technology Stack

This project was developed using **Python** in **Google Colab**, supported by industry-standard libraries for data preprocessing, statistical analysis, and visualization. These technologies enabled efficient data loading, cleaning, transformation, exploratory analysis, and the creation of meaningful visualizations to derive actionable business insights from the logistics dataset.

Technology Stack

Technology / Tool	Purpose
Python	Primary programming language used for data processing, statistical analysis, and visualization.
Google Colab	Cloud-based development environment for writing, executing, and sharing Python notebooks.
Pandas	Data loading, cleaning, preprocessing, manipulation, and exploratory data analysis.
NumPy	Numerical computations, mathematical operations, and efficient array processing.
Matplotlib	Creation of static charts and graphical visualizations for data interpretation.
Seaborn	Advanced statistical visualizations with improved aesthetics and analytical insights.
Plotly	Development of interactive and dynamic visualizations for exploratory analysis.
Jupyter Notebook (.ipynb)	Documentation and implementation of the complete Exploratory Data Analysis workflow.

Technology Workflow



The integration of these technologies provided a robust analytical environment for processing large volumes of logistics data, performing statistical analysis, and creating informative visualizations. This technology stack facilitated the identification of operational trends, cost patterns, shipment behavior, and supply chain inefficiencies, ultimately supporting data-driven decision-making and logistics optimization.

10. Exploratory Data Analysis (EDA) Methodology

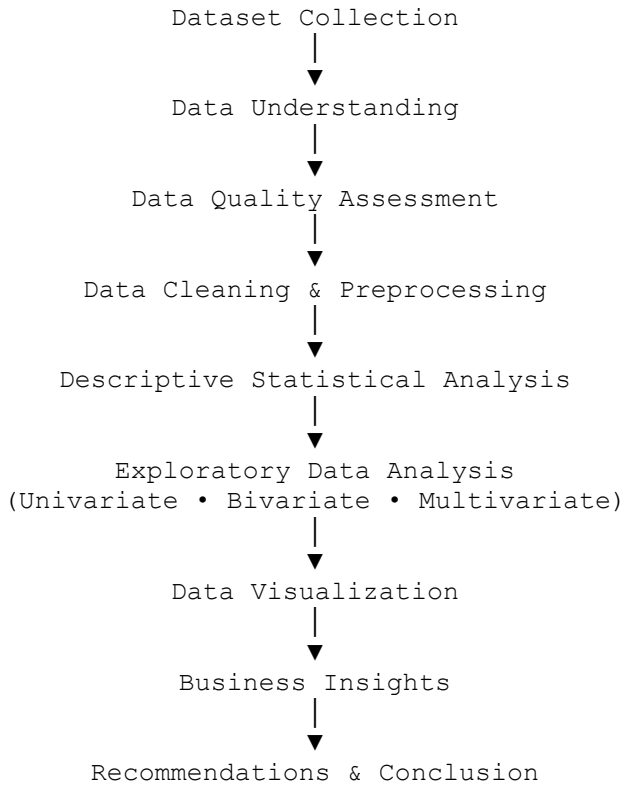
The **Exploratory Data Analysis (EDA)** methodology adopted in this project follows a systematic and structured approach to transform raw logistics data into meaningful business insights. The process begins with **data understanding**, where the dataset is examined to identify its dimensions, data types, variables, and overall structure. A comprehensive data quality assessment is then conducted to detect missing values, duplicate records, inconsistent entries, and formatting issues that may affect the accuracy of the analysis.

The next phase involves **data cleaning and preprocessing**, including handling missing values, converting date fields into appropriate formats, correcting inconsistent data entries, removing unnecessary spaces, and preparing the dataset for analysis. Once the data is cleaned, **descriptive statistical analysis** is performed to summarize key numerical variables such as shipment weight, freight cost, line item value, insurance cost, and shipment quantity using appropriate statistical measures.

The exploratory analysis is subsequently divided into **Univariate, Bivariate, and Multivariate Analysis** to gain a comprehensive understanding of the dataset. Univariate analysis examines the distribution of individual variables, bivariate analysis explores relationships between two variables, and multivariate analysis investigates interactions among multiple features. A variety of visualization techniques—including bar charts, histograms, box plots, scatter plots, violin plots, bubble charts, correlation heatmaps, pair plots, and lollipop charts—are utilized to identify shipment trends, transportation patterns, cost drivers, regional performance, and operational bottlenecks.

Finally, the analytical findings are interpreted to generate actionable business insights and practical recommendations aimed at improving logistics efficiency, optimizing transportation planning, reducing operational costs, and supporting data-driven strategic decision-making. This structured methodology ensures that the analysis is reliable, reproducible, and aligned with the project's objective of enhancing FedEx logistics operations through evidence-based insights.

EDA Methodology Workflow



11. Data Understanding

Data Understanding is the initial and one of the most critical phases of **Exploratory Data Analysis (EDA)**, as it provides a clear understanding of the dataset's structure, quality, and overall characteristics before detailed analysis begins. In this project, the **SCMS Delivery History Dataset** was examined to understand its dimensions, data types, variables, and data quality. The dataset contains historical shipment records related to logistics and supply chain operations, including shipment information, transportation modes, freight costs, delivery schedules, product details, vendors, manufacturing sites, destination countries, and insurance values.

The dataset comprises **10,324 records** and **33 features**, offering a comprehensive representation of logistics activities. Among these features, **7 are numerical** and **26 are categorical**, allowing both statistical and categorical analyses. Data quality assessment revealed **no duplicate records**, indicating good data consistency. However, **2,383 missing values** were identified, primarily in the **Dosage**, **Shipment Mode**, and **Line Item Insurance (USD)** columns. The dataset occupies approximately **16.04 MB** of memory, making it suitable for efficient processing and analysis.

Before performing visualization and statistical analysis, an initial exploration was conducted by inspecting the first few records, examining feature names and data types, identifying unique values, evaluating missing data, and generating descriptive statistics. This preliminary assessment provided a better understanding of the dataset and helped identify data quality issues that required preprocessing before conducting Exploratory Data Analysis.

Dataset Overview

Parameter	Value
Dataset Name	SCMS Delivery History Dataset
Total Records	10,324
Total Features	33
Numerical Features	7
Categorical Features	26
Duplicate Records	0
Missing Values	2,383
Memory Usage	16.04 MB

Key Observations

- The dataset contains **10,324 shipment records** and **33 business-related attributes**, providing comprehensive coverage of logistics operations.
- No duplicate records were identified, indicating a high level of data consistency.
- Missing values are concentrated mainly in the **Dosage (16.82%)**, **Shipment Mode (3.49%)**, and **Line Item Insurance (USD) (2.78%)** columns, requiring appropriate preprocessing.
- The majority of the variables are categorical, representing shipment characteristics, vendors, manufacturing sites, product information, and destination countries.
- Key numerical variables, including **Freight Cost (USD)**, **Weight (Kilograms)**, **Line Item Value**, **Line Item Quantity**, and **Insurance Cost**, are critical for evaluating transportation costs and logistics performance.
- The dataset provides sufficient information to analyze shipment trends, transportation efficiency, regional distribution, cost behavior, and overall supply chain performance through Exploratory Data Analysis (EDA).

12. Data Cleaning & Data Preparation

Data Cleaning and Data Preparation were performed to improve the quality, consistency, and reliability of the dataset before conducting **Exploratory Data Analysis (EDA)**. The objective of this phase was to prepare the raw shipment data for meaningful statistical analysis and visualization by addressing formatting inconsistencies, validating data quality, and ensuring the dataset was structured appropriately for analysis.

To preserve the original dataset, a separate copy was created before any preprocessing activities were performed. Column names were standardized by removing leading and trailing spaces, while text-based fields were cleaned to eliminate unnecessary whitespace and maintain consistency across categorical variables. Delivery-related date columns were converted into the **datetime** format, enabling accurate time-based analysis and comparison of shipment schedules.

A comprehensive data quality assessment was conducted to identify duplicate records and missing values. No duplicate records were found, confirming the integrity of the dataset. Missing values were analyzed and retained where appropriate, as they represented legitimate business information rather than data entry errors. Finally, the dataframe index was reset to maintain a clean and sequential structure after preprocessing.

After completing all preprocessing steps, the dataset retained its original size of **10,324 records** and **33 features**, indicating that no observations were removed during the cleaning process. The prepared dataset provided a reliable foundation for subsequent statistical analysis, visualization, and business insight generation.

Data Preparation Summary

Data Cleaning Activity	Description
Dataset Copy	Created a copy of the original dataset to preserve raw data.
Column Name Standardization	Removed leading and trailing spaces from column names for consistency.
Text Cleaning	Eliminated unnecessary whitespace from text-based columns.
Date Conversion	Converted delivery-related date columns into the datetime format.
Duplicate Record Check	Verified that no duplicate records existed in the dataset.
Missing Value Assessment	Identified and documented missing values for further analysis without deleting records.
Index Reset	Reset the dataframe index after preprocessing to maintain consistency.

Data Cleaning Results

Parameter	Value
Original Records	10,324
Final Records	10,324
Duplicates Removed	0
Remaining Missing Values	2,383
Total Features	33

Key Outcomes

- The dataset was successfully cleaned, standardized, and prepared for analysis.
- No duplicate records were identified, ensuring high data consistency.
- Date fields were converted into the appropriate format to support temporal analysis.
- Text formatting inconsistencies were removed, improving the quality of categorical data.
- Missing values were identified, documented, and retained where appropriate to preserve valuable business information.
- The cleaned dataset provided a reliable foundation for statistical analysis, data visualization, and the generation of meaningful business insights.

13. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a systematic process of examining, summarizing, and visualizing data to identify patterns, trends, relationships, and anomalies before applying advanced analytical techniques. In this project, EDA was conducted on the **FedEx Supply Chain Logistics Dataset** to gain comprehensive insights into shipment operations, transportation performance, freight costs, customer distribution, and overall supply chain efficiency.

The analysis was performed using **Python** in **Google Colab**, utilizing libraries such as **Pandas**, **NumPy**, **Matplotlib**, **Seaborn**, and **Plotly**. A total of **15 business-oriented visualizations** were developed to explore various aspects of the logistics dataset. The analysis follows the **UBM (Univariate–Bivariate–Multivariate)** framework, which first examines individual variables, then analyzes relationships between pairs of variables, and finally investigates interactions among multiple variables simultaneously.

The insights derived from this analysis help identify operational trends, cost drivers, shipment patterns, regional performance, and opportunities for improving logistics efficiency and supporting data-driven decision-making.

13.1 Univariate Analysis

Overview

Univariate Analysis focuses on analyzing **a single variable at a time** to understand its distribution, frequency, central tendency, variability, and overall characteristics. It provides a fundamental understanding of each feature individually and helps identify dominant categories, skewness, outliers, and data quality issues.

In this project, univariate analysis was performed on key logistics variables, including shipment destinations, shipment modes, vendors, manufacturing sites, freight costs, shipment weights, and product-related attributes.

Charts Included

Chart No.	Visualization	Purpose
Chart 1	Top 10 Destination Countries	Identify countries receiving the highest number of shipments.
Chart 2	Shipment Mode Distribution	Analyze the utilization of different shipment modes.
Chart 3	Top 10 Vendors	Identify major vendors contributing to shipment volume.
Chart 4	Top Manufacturing Sites	Examine manufacturing sites with the highest shipment activity.
Chart 5	Freight Cost Distribution (Histogram)	Analyze the distribution and variability of freight costs.

Chart 6	Shipment Weight Distribution	Examine shipment weight patterns and detect skewness or outliers.
Chart 10	Lollipop Chart – Destination Countries	Compare shipment frequencies across major destination countries.
Chart 13	Donut Chart – Shipment Mode	Visualize the percentage share of each shipment mode.

Key Findings

- Air transportation accounts for the largest share of total shipments, indicating its importance in the logistics network.
- **South Africa**, **Nigeria**, and **Côte d'Ivoire** are among the leading shipment destinations.
- Freight cost and shipment weight exhibit positively skewed distributions with several high-value outliers.
- A relatively small number of vendors and manufacturing sites contribute to a significant proportion of total shipments.
- Shipment volumes are concentrated within a limited number of destination countries, highlighting key operational regions.

13.2 Bivariate Analysis

Overview

Bivariate Analysis examines the relationship between **two variables** to determine how changes in one variable influence another. This approach helps identify correlations, comparative trends, and associations between numerical and categorical variables.

In this project, bivariate analysis was conducted to investigate relationships between shipment mode, freight cost, shipment weight, destination country, and transportation expenses.

Charts Included

Chart No.	Visualization	Purpose
Chart 7	Shipment Mode vs Freight Cost (Box Plot)	Compare freight costs across different shipment modes.
Chart 8	Shipment Weight vs Freight Cost (Scatter Plot)	Examine the relationship between shipment weight and freight cost.
Chart 9	Average Freight Cost by Country	Compare average transportation costs across destination countries.
Chart 11	Shipment Weight by Shipment Mode (Violin Plot)	Analyze shipment weight distributions across transportation modes.

Key Findings

- Freight cost generally increases with shipment weight, indicating a positive relationship.
- Air transportation incurs higher freight costs compared to other shipment modes due to its premium delivery services.
- Certain destination countries exhibit significantly higher average freight costs, suggesting regional cost variations.
- Shipment mode is a key factor influencing transportation expenses and delivery strategies.
- Heavy shipments contribute disproportionately to overall logistics costs.

13.3 Multivariate Analysis

Overview

Multivariate Analysis examines the interaction among **three or more variables** simultaneously to uncover complex relationships that cannot be identified through univariate or bivariate analysis alone. This approach provides a comprehensive understanding of the combined effects of multiple business variables on logistics operations.

The analysis explores relationships among shipment weight, freight cost, shipment quantity, insurance cost, shipment mode, and line item value to identify operational patterns and business opportunities.

Charts Included

Chart No.	Visualization	Purpose
Chart 12	Bubble Chart	Analyze the combined relationship among shipment weight, freight cost, shipment mode, and shipment quantity.
Chart 14	Correlation Heatmap	Measure correlations among key numerical logistics variables.
Chart 15	Pair Plot	Examine relationships among multiple numerical variables simultaneously.

Key Findings

- Shipment weight demonstrates a moderate positive correlation with freight cost.
- Line Item Value has a strong positive relationship with Line Item Insurance, indicating higher insurance requirements for valuable shipments.
- Line Item Quantity positively influences overall shipment value and transportation requirements.

- Freight cost is influenced not only by shipment weight but also by shipment mode and shipment quantity.
- The combined analysis highlights opportunities for transportation cost optimization, operational planning, and resource allocation.

Summary of Exploratory Data Analysis

The Exploratory Data Analysis successfully transformed raw logistics data into meaningful business intelligence. **Univariate Analysis** provided insights into the distribution and characteristics of individual variables, **Bivariate Analysis** revealed relationships affecting transportation costs and shipment performance, while **Multivariate Analysis** uncovered complex interactions among multiple logistics variables. Collectively, these findings provide a comprehensive understanding of FedEx's logistics operations and establish a strong foundation for the statistical analysis, business insights, and strategic recommendations presented in the subsequent sections of this report.

14. Statistical Summary

Statistical analysis was conducted on the numerical variables in the **FedEx Supply Chain Logistics Dataset** to understand their central tendency, dispersion, and overall distribution. This analysis provides a quantitative overview of the dataset and helps identify variability, outliers, and potential business trends before performing advanced visual and relationship-based analyses.

The dataset contains **7 numerical variables**: **ID**, **Unit of Measure (Per Pack)**, **Line Item Quantity**, **Line Item Value (USD)**, **Pack Price (USD)**, **Unit Price (USD)**, and **Line Item Insurance (USD)**. These variables represent key operational aspects of logistics, including shipment quantity, product pricing, shipment value, and insurance coverage.

The statistical summary indicates considerable variation across the numerical variables. **Line Item Quantity** has an average of **18,332.53 units**, with values ranging from **1** to **619,999**, demonstrating that the dataset includes both small and large-scale shipments. Similarly, **Line Item Value (USD)** has a mean of **USD 157,650.57**, while the maximum value reaches **USD 5,951,990.40**, highlighting substantial differences in shipment values.

The average **Pack Price (USD)** is **21.91**, whereas the average **Unit Price (USD)** is **0.61**, indicating that products are generally transported in bulk quantities. Additionally, **Line Item Insurance (USD)** has an average value of **USD 240.12**, with a maximum of **USD 7,708.44**, suggesting that higher-value shipments require greater insurance coverage and risk protection.

Overall, the statistical analysis reveals that most numerical variables exhibit **positively skewed distributions** with several extreme outliers. These characteristics emphasize the importance of using visualization techniques such as histograms, box plots, scatter plots, and

correlation heatmaps to better understand shipment behavior, transportation costs, and logistics performance.

Statistical Summary of Numerical Variables

Variable	Count	Mean	Standard Deviation	Minimum	Median	Maximum
ID	10,324	51,098.97	31,944.33	1	57,540.50	86,823
Unit of Measure (Per Pack)	10,324	77.99	76.58	1	60	1,000
Line Item Quantity	10,324	18,332.53	40,035.30	1	3,000	619,999
Line Item Value (USD)	10,324	157,650.57	345,292.07	0.00	30,471.47	5,951,990.40
Pack Price (USD)	10,324	21.91	45.61	0.00	9.30	1,345.64
Unit Price (USD)	10,324	0.61	3.28	0.00	0.16	238.65
Line Item Insurance (USD)	10,037	240.12	500.19	0.00	47.04	7,708.44

Key Statistical Observations

- The dataset contains **7 numerical variables**, providing quantitative insights into shipment operations, pricing, and insurance.
- **Line Item Quantity** and **Line Item Value (USD)** exhibit high variability, reflecting significant differences in shipment sizes and financial values.
- **Pack Price (USD)** and **Unit Price (USD)** display positively skewed distributions, indicating that most products are moderately priced while a few premium products have substantially higher prices.
- **Line Item Insurance (USD)** increases with shipment value, highlighting the importance of insurance coverage for high-value shipments.
- Most numerical variables contain **extreme outliers**, demonstrating the presence of exceptionally large shipments and high-value transactions.
- The statistical findings establish a strong quantitative foundation for understanding shipment characteristics, pricing behavior, transportation costs, and overall logistics performance, supporting the visual and analytical insights presented in subsequent sections.

15. Business Insights

The **Exploratory Data Analysis (EDA)** of the **FedEx Supply Chain Logistics Dataset** generated valuable insights into shipment operations, transportation performance, freight costs, regional distribution, and overall supply chain efficiency. These findings provide a data-driven foundation for improving operational performance, optimizing logistics strategies, reducing transportation costs, and enhancing customer satisfaction.

Key Business Insights

1. High Shipment Concentration in Key Markets

The analysis revealed that a substantial proportion of shipments are concentrated in **South Africa, Nigeria, and Côte d'Ivoire**. These countries represent major logistics hubs within the distribution network and require continued investment in transportation infrastructure, inventory planning, and resource allocation to sustain operational efficiency.

2. Air Transportation Dominates Logistics Operations

Air transport is the most frequently used shipment mode, reflecting FedEx's commitment to fast and reliable delivery services. While this approach improves delivery speed and customer satisfaction, it also contributes to higher transportation costs. Evaluating alternative shipment modes for non-time-sensitive deliveries could improve overall cost efficiency.

3. Shipment Weight is a Key Driver of Freight Cost

The analysis identified a **moderate positive correlation** between shipment weight and freight cost. As shipment weight increases, transportation expenses also tend to rise. This finding highlights the importance of load optimization, packaging efficiency, and shipment consolidation to reduce logistics costs.

4. Transportation Mode Significantly Influences Shipping Expenses

Freight costs vary considerably across different shipment modes. Air transportation generally incurs the highest costs compared with other available modes, indicating opportunities to optimize transportation planning by selecting more economical shipping methods where delivery urgency permits.

5. Shipment Value Directly Affects Insurance Cost

A strong positive relationship exists between **Line Item Value (USD)** and **Line Item Insurance (USD)**. High-value shipments require greater insurance coverage, emphasizing the need for effective risk management, secure transportation, and appropriate insurance planning for valuable goods.

6. Strategic Vendors and Manufacturing Sites Drive Operations

A relatively small number of vendors and manufacturing sites account for a significant share of shipment activity. Strengthening partnerships with these key suppliers and optimizing their operational performance can improve procurement efficiency, inventory availability, and supply chain reliability.

7. Regional Variations Affect Logistics Performance

Average freight costs vary significantly across destination countries, suggesting differences in transportation infrastructure, shipping distance, and operational complexity. These regional variations present opportunities to optimize shipping routes, negotiate carrier contracts, and improve regional logistics planning.

8. Presence of High-Value and Large-Scale Shipments

The dataset contains several outlier shipments with exceptionally high weights, freight costs, and shipment values. These specialized shipments may require dedicated transportation planning, enhanced monitoring, and additional handling procedures to minimize operational risks and unexpected expenses.

9. Multiple Business Variables Influence Logistics Costs

Correlation analysis indicates that shipment quantity, shipment weight, line item value, insurance cost, and freight cost are interrelated. Considering these variables collectively enables more effective transportation planning, cost estimation, and resource allocation.

10. High Data Quality Supports Reliable Decision-Making

The dataset contains **no duplicate records** and only a limited number of missing values, indicating good overall data quality. This ensures that the analytical findings are reliable and suitable for supporting evidence-based business decisions.

Summary of Business Insights

Business Area	Key Insight	Business Impact
Shipment Distribution	Shipments are concentrated in a few high-demand countries.	Prioritize logistics infrastructure and resource allocation in strategic regions.
Transportation Mode	Air transport is the dominant shipment method.	Evaluate alternative transportation modes to improve cost efficiency.
Freight Cost	Transportation cost increases with shipment weight.	Optimize packaging, shipment consolidation, and load planning.
Regional Performance	Freight costs vary significantly across destination countries.	Improve route planning, carrier selection, and regional logistics strategies.
Vendor Performance	A limited number of vendors account for most shipments.	Strengthen strategic supplier relationships and procurement efficiency.

Insurance Management	High-value shipments require greater insurance coverage.	Enhance risk management and shipment security practices.
Operational Outliers	Large and high-value shipments generate exceptionally high logistics costs.	Implement specialized handling and transportation strategies.
Data Quality	No duplicate records and manageable missing values.	Supports accurate analysis and reliable data-driven decision-making.

Overall Business Perspective

The analysis demonstrates that **shipment weight, transportation mode, freight cost, destination country, shipment quantity, and line item value** are the primary factors influencing logistics performance and operational efficiency. By leveraging these insights, FedEx can optimize transportation planning, improve resource utilization, reduce logistics costs, strengthen supply chain resilience, and enhance customer satisfaction through data-driven decision-making. These findings also provide a strong foundation for future predictive analytics, demand forecasting, and logistics optimization initiatives.

16. Business Recommendations

Based on the findings of the **Exploratory Data Analysis (EDA)**, the following recommendations are proposed to enhance FedEx's logistics operations, improve transportation efficiency, reduce operational costs, and strengthen overall supply chain performance. These recommendations are derived from the shipment patterns, cost relationships, and operational trends identified throughout the analysis.

1. Optimize High-Volume Shipping Routes

The analysis identified **South Africa, Nigeria, and Côte d'Ivoire** as major shipment destinations. FedEx should prioritize these high-demand markets by optimizing transportation routes, expanding regional distribution capabilities, and improving warehouse capacity to enhance delivery efficiency and reduce transit times.

2. Improve Freight Cost Efficiency through Shipment Consolidation

Since freight cost increases with shipment weight, the company should implement shipment consolidation strategies, optimize packaging practices, and maximize vehicle capacity utilization. Consolidating compatible shipments can reduce transportation expenses while maintaining service quality and delivery performance.

3. Optimize Transportation Mode Selection

Air transportation is the most frequently used shipment mode but also represents the highest transportation cost. FedEx should evaluate shipment urgency and shift non-time-sensitive deliveries to more cost-effective transportation modes, such as ocean or ground transport, whenever operationally feasible.

4. Strengthen Vendor and Supplier Management

A relatively small number of vendors and manufacturing sites contribute to a significant share of shipment activity. Establishing stronger strategic partnerships, implementing regular vendor performance evaluations, and negotiating long-term agreements can improve procurement efficiency, supply reliability, and operational resilience.

5. Enhance Risk Management for High-Value Shipments

The strong relationship between shipment value and insurance cost highlights the importance of effective risk management. FedEx should strengthen packaging standards, improve shipment monitoring, and adopt secure transportation practices to minimize product damage, reduce insurance claims, and protect valuable goods throughout the delivery process.

6. Improve Regional Logistics Performance

Significant variations in freight costs across destination countries indicate opportunities for regional optimization. The company should analyze high-cost regions, optimize transportation routes, negotiate competitive carrier agreements, and establish regional distribution centers where economically viable.

7. Monitor High-Cost and Large-Scale Shipments

Exceptional shipments with unusually high weights, values, or freight costs should be managed through dedicated operational planning, specialized handling procedures, and continuous cost monitoring. This approach will help reduce operational risks and improve resource allocation.

8. Strengthen Data Quality Management

Although the dataset demonstrates good overall quality, missing values were identified in a few business variables. Implementing standardized data collection procedures, automated validation rules, and regular data quality audits will improve reporting accuracy and support more reliable business analytics.

9. Adopt Predictive Analytics for Decision Support

The relationships identified among shipment weight, freight cost, shipment quantity, insurance cost, and shipment value provide an excellent foundation for predictive analytics. Machine learning models can be developed to forecast freight costs, predict shipment demand, optimize inventory planning, and estimate delivery performance, enabling proactive and data-driven decision-making.

10. Develop Interactive Business Intelligence Dashboards

FedEx should implement interactive dashboards to monitor key performance indicators (KPIs), including shipment volume, freight cost, delivery performance, transportation mode utilization, regional operations, and vendor performance. Real-time monitoring will improve operational visibility, accelerate decision-making, and support continuous performance improvement.

Summary of Recommendations

Recommendation	Expected Business Benefit
Optimize high-volume shipping routes	Improve delivery efficiency and reduce transportation time.
Consolidate shipments and optimize packaging	Reduce freight costs and improve vehicle utilization.
Optimize transportation mode selection	Lower logistics costs while maintaining service quality.
Strengthen vendor and supplier management	Improve procurement efficiency and supply chain reliability.
Enhance risk and insurance management	Protect high-value shipments and reduce financial losses.
Improve regional logistics planning	Reduce transportation costs in high-expense regions.
Monitor high-cost shipments	Minimize operational risks and optimize resource utilization.
Strengthen data quality management	Improve the accuracy and reliability of business analytics.
Implement predictive analytics	Enhance forecasting, planning, and strategic decision-making.
Develop interactive dashboards	Enable real-time performance monitoring and continuous operational improvement.

Overall Recommendation

The successful implementation of these recommendations will enable FedEx to optimize transportation planning, improve operational efficiency, reduce logistics costs, strengthen supplier relationships, and enhance customer satisfaction. Furthermore, integrating predictive analytics, business intelligence dashboards, and data-driven decision-making into logistics operations will improve supply chain visibility, support proactive management, and provide a sustainable competitive advantage in the rapidly evolving global logistics and transportation industry.

17. Challenges Faced

During the **Exploratory Data Analysis (EDA)** of the **FedEx Supply Chain Logistics Dataset**, several data quality and analytical challenges were encountered. Addressing these challenges was essential to ensure the accuracy, consistency, and reliability of the analysis. Appropriate data preprocessing and validation techniques were applied to prepare the dataset for meaningful statistical analysis and visualization.

1. Missing Values

The dataset contained **2,383 missing values**, primarily in the **Dosage**, **Shipment Mode**, and **Line Item Insurance (USD)** columns. These missing entries required careful evaluation to determine whether they represented genuine business scenarios or incomplete records. Instead of indiscriminately removing observations, the missing values were documented and handled appropriately to preserve valuable business information.

2. Inconsistent Data Types

Certain columns expected to contain numerical values, such as **Weight (Kilograms)** and **Freight Cost (USD)**, included non-numeric text entries such as *"Freight Included in Commodity Cost"* and *"See ASN-93 (ID#:1281)"*. These inconsistencies prevented direct numerical analysis and required data type conversion with appropriate handling of invalid values.

3. Date Format Standardization

Delivery-related date fields were initially stored as text rather than date objects. These columns were converted into the **datetime** format to enable accurate chronological analysis, delivery comparisons, and future time-series investigations.

4. Presence of Outliers

Several numerical variables, including **Shipment Weight**, **Freight Cost**, and **Line Item Value**, contained extreme observations. These outliers significantly influenced statistical measures and data distributions. Appropriate visualization techniques and analytical methods were used to understand their impact without removing potentially important business records.

5. High Cardinality of Categorical Variables

Variables such as **Country**, **Vendor**, and **Manufacturing Site** contained a large number of unique categories, making comprehensive visualization difficult. To improve readability and interpretation, the analysis focused on the most significant categories, such as the top 10 or top 15 contributors based on shipment volume.

6. Skewed Data Distributions

Most numerical variables exhibited **positively skewed distributions**, with many low-value observations and a relatively small number of extremely large values. Histograms, box plots, and distribution analysis were used to better understand these patterns and identify unusual observations.

7. Extensive Data Preparation Requirements

Before performing Exploratory Data Analysis, several preprocessing activities were required, including cleaning column names, removing unnecessary whitespace, converting data types, handling missing values, validating records, and preparing the dataset for visualization. These steps ensured that the analysis was accurate, consistent, and reproducible.

Summary of Challenges

Challenge	Impact	Solution Applied
Missing values	Incomplete information in selected variables	Identified, documented, and appropriately handled missing values while preserving valid business records.
Mixed data types	Errors during numerical analysis	Converted text-based numeric fields into appropriate numerical formats and handled invalid entries.
Date formatting	Limited ability to perform temporal analysis	Converted date columns into the datetime format.
Outliers	Distorted statistical summaries and visualizations	Used appropriate visualization techniques and analytical methods to interpret extreme values.
High-cardinality categorical variables	Complex and cluttered visualizations	Focused on the most significant categories for clearer interpretation.
Positively skewed distributions	Uneven representation of numerical variables	Applied histograms, box plots, and distribution analysis to evaluate data behavior.
Data preprocessing requirements	Increased complexity of data preparation	Performed comprehensive data cleaning and preprocessing before analysis.

Overall Assessment

Despite these challenges, appropriate data cleaning, preprocessing, and visualization techniques ensured that the dataset was suitable for comprehensive analysis. The preprocessing process improved data quality, enhanced analytical accuracy, and enabled the generation of reliable business insights. Consequently, the findings of this project provide a dependable foundation for data-driven decision-making and logistics optimization within FedEx's supply chain operations.

18. Limitations

While this **Exploratory Data Analysis (EDA)** provides meaningful insights into FedEx logistics operations, certain limitations must be acknowledged. These constraints are primarily related to data quality, analytical scope, and methodological boundaries, and should be considered when interpreting the results.

1. Missing and Incomplete Data

The dataset contains **2,383 missing values** across key variables such as **Dosage, Shipment Mode, and Line Item Insurance (USD)**. Although these missing values were analyzed and retained where appropriate, their presence may have a minor impact on the precision of certain insights.

2. Data Inconsistencies

Some numerical fields, including **Weight (Kilograms)** and **Freight Cost (USD)**, contain non-standard textual entries such as *"Freight Included in Commodity Cost"*. These inconsistencies required conversion and filtering during preprocessing, which may have led to slight data exclusion in certain cases.

3. Influence of Outliers

Several variables, including **shipment weight, freight cost, and line item value**, contain extreme values. Although these outliers were managed using appropriate visualization techniques and statistical handling methods, they may still influence overall trends, averages, and variability measures.

4. Limited Temporal Analysis

Although the dataset includes date-related fields, the analysis primarily focuses on exploratory patterns rather than time-series modeling. As a result, deeper insights into seasonal trends, monthly variations, and long-term forecasting were not explored.

5. Aggregated-Level Analysis

The analysis is conducted at an aggregated level (e.g., country, vendor, shipment mode), which may obscure finer operational details at the individual transaction or shipment level.

6. Absence of Predictive Modeling

This project is limited to **Exploratory Data Analysis (EDA)** and does not incorporate machine learning or predictive analytics techniques. Therefore, forecasting of freight costs, shipment delays, or demand patterns is not included.

7. Static Analytical Approach

The analysis is based on historical data and static visualizations. It does not include real-time data processing or interactive dashboards, limiting its use for continuous monitoring and live decision-making.

8. Lack of External Data Integration

The dataset is analyzed in isolation without incorporating external influencing factors such as fuel price fluctuations, weather conditions, geopolitical events, or market demand changes, which may impact logistics performance in real-world scenarios.

Summary of Limitations

Limitation	Impact
Missing values	May slightly affect the accuracy of analytical results
Data inconsistencies	Requires preprocessing and may reduce data completeness
Outliers	Can influence statistical measures and distribution patterns
Limited time-series analysis	Restricts understanding of seasonal and temporal trends
Aggregated-level analysis	May hide granular operational insights
No predictive modeling	Limits forecasting and future trend estimation
Static analysis approach	Does not support real-time monitoring
No external data integration	Reduces contextual completeness of insights

Overall Note

Despite these limitations, the dataset remains highly suitable for exploratory analysis and provides valuable insights into logistics performance, shipment behavior, and cost structures. These limitations also highlight potential areas for future enhancement, including predictive analytics, real-time dashboards, and integration of external data sources to support more advanced decision-making.

19. Future Scope

The **Exploratory Data Analysis (EDA)** conducted in this project establishes a strong foundation for future research and the development of advanced analytics solutions in logistics and supply chain management. While the present study focuses on understanding historical shipment data and identifying operational trends, numerous opportunities exist to extend the analysis through predictive modeling, artificial intelligence, and real-time decision support systems.

1. Freight Cost Prediction

Machine learning algorithms such as **Linear Regression**, **Random Forest**, **Gradient Boosting**, and **XGBoost** can be developed to predict freight costs based on shipment weight, destination, transportation mode, shipment quantity, and product characteristics. Accurate cost prediction can support transportation budgeting, pricing strategies, and logistics planning.

2. Delivery Delay Prediction

Historical shipment and delivery records can be utilized to build predictive models capable of identifying shipments that are likely to experience delays. Early detection of potential delays would enable proactive planning, improve service reliability, and enhance customer satisfaction.

3. Demand Forecasting

Time-series forecasting techniques, including **ARIMA**, **Prophet**, and **Long Short-Term Memory (LSTM)** neural networks, can be implemented to forecast future shipment volumes and demand patterns. These forecasts would support inventory management, workforce planning, transportation capacity allocation, and resource optimization.

4. Intelligent Route Optimization

Integrating geographical information with optimization algorithms can help identify the most efficient transportation routes based on distance, delivery schedules, traffic conditions, and transportation costs. This approach can reduce fuel consumption, shorten delivery times, and improve overall logistics efficiency.

5. Real-Time Business Intelligence Dashboards

Interactive dashboards developed using **Power BI**, **Tableau**, **Plotly Dash**, or similar business intelligence platforms can provide real-time monitoring of key performance indicators (KPIs), including shipment volume, freight cost, delivery performance, transportation mode utilization, regional operations, and vendor performance. Such dashboards would improve operational visibility and support faster managerial decision-making.

6. Integration of External Data Sources

Future studies can enhance analytical accuracy by incorporating external data such as weather conditions, fuel prices, traffic congestion, geopolitical events, exchange rates, and economic indicators. Integrating these variables would provide a more comprehensive understanding of the factors influencing logistics performance.

7. Advanced Customer and Supplier Analytics

Additional analytical techniques can be applied to perform customer segmentation, vendor performance evaluation, supplier risk assessment, and procurement optimization. These analyses can strengthen supplier relationships, improve procurement strategies, and increase overall supply chain resilience.

8. Artificial Intelligence for Supply Chain Optimization

Artificial Intelligence (AI) and Deep Learning techniques can be integrated to automate shipment planning, inventory optimization, transportation scheduling, route selection, and demand forecasting. AI-driven decision support systems can improve operational efficiency, reduce manual intervention, and enhance strategic planning across the supply chain.

Summary of Future Scope

Future Enhancement	Expected Business Benefit
Freight cost prediction	Improve transportation budgeting and cost estimation.
Delivery delay prediction	Reduce delays and enhance customer satisfaction.
Demand forecasting	Optimize inventory, workforce, and transportation planning.
Intelligent route optimization	Reduce delivery time, fuel consumption, and logistics costs.
Real-time business dashboards	Enable continuous monitoring of logistics performance and KPIs.
External data integration	Improve forecasting accuracy and operational planning.
Customer and supplier analytics	Strengthen supplier relationships and supply chain efficiency.
AI-based supply chain optimization	Enhance automation, operational efficiency, and strategic decision-making.

Overall Future Direction

The insights generated through this Exploratory Data Analysis provide a valuable foundation for developing intelligent, data-driven logistics solutions. By integrating **machine learning, predictive analytics, business intelligence, real-time monitoring, and artificial intelligence**, FedEx can further optimize transportation planning, improve delivery performance, reduce operational costs, strengthen supply chain resilience, and enhance customer satisfaction. These advancements will enable more informed decision-making and support the development of a smarter, more efficient, and future-ready global logistics network.

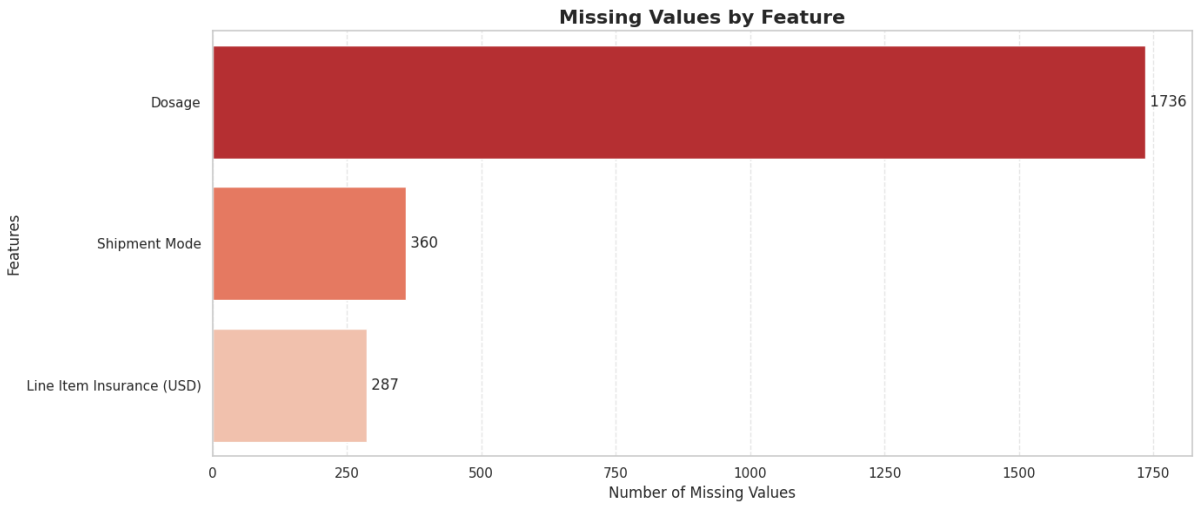
20. Graph Visualizations (Screenshots)

This section presents the graphical visualizations created during the **Exploratory Data Analysis (EDA)** of the **FedEx Supply Chain Logistics Dataset**. These visualizations provide a clear understanding of shipment distribution, transportation performance, freight costs, supplier contribution, regional operations, and relationships among key logistics variables.

Each figure has been included to support the analytical findings and demonstrate how data visualization can transform raw logistics data into meaningful business insights for strategic decision-making.

Figure: Missing Values by Feature

Screenshot



Description

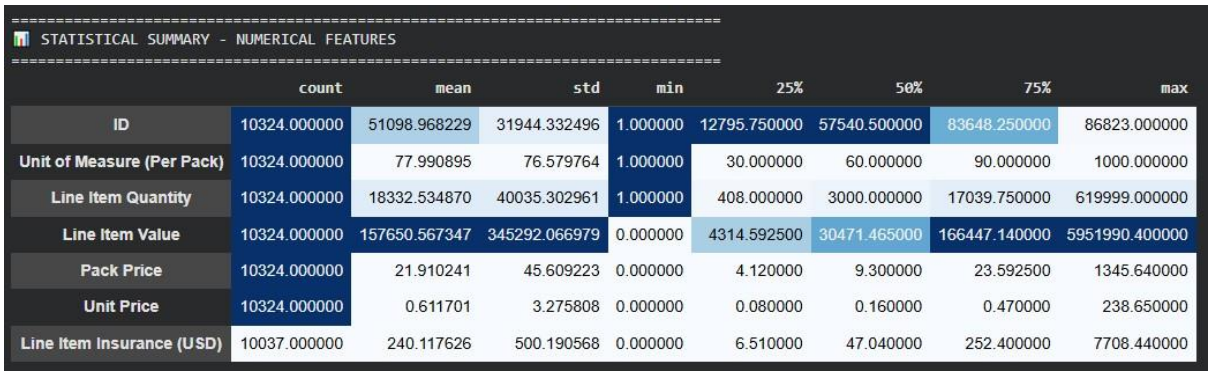
This bar chart illustrates the number of missing values present in each feature of the dataset. It highlights columns containing incomplete information before data preprocessing.

Business Interpretation

- The **Dosage** column contains the highest number of missing values, followed by **Shipment Mode** and **Line Item Insurance (USD)**.
- Most features contain complete information, indicating that the dataset is generally of good quality.
- Understanding missing data helps determine appropriate preprocessing techniques and ensures reliable analytical results.

Figure: Statistical Summary of Numerical Features

Screenshot



STATISTICAL SUMMARY - NUMERICAL FEATURES

	count	mean	std	min	25%	50%	75%	max
ID	10324.000000	51098.968229	31944.332496	1.000000	12795.750000	57540.500000	83648.250000	86823.000000
Unit of Measure (Per Pack)	10324.000000	77.990895	76.579764	1.000000	30.000000	60.000000	90.000000	1000.000000
Line Item Quantity	10324.000000	18332.534870	40035.302961	1.000000	408.000000	3000.000000	17039.750000	619999.000000
Line Item Value	10324.000000	157650.567347	345292.066979	0.000000	4314.592500	30471.465000	166447.140000	5951990.400000
Pack Price	10324.000000	21.910241	45.609223	0.000000	4.120000	9.300000	23.592500	1345.640000
Unit Price	10324.000000	0.611701	3.275808	0.000000	0.080000	0.160000	0.470000	238.650000
Line Item Insurance (USD)	10037.000000	240.117626	500.190568	0.000000	6.510000	47.040000	252.400000	7708.440000

Description

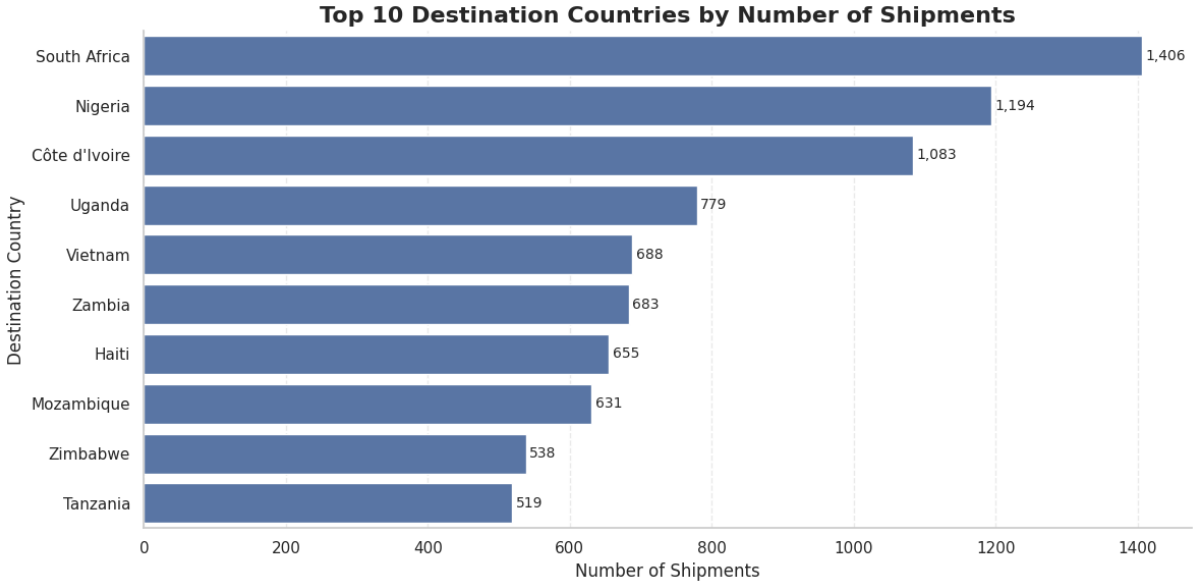
This statistical summary table provides descriptive statistics for all numerical variables, including count, mean, standard deviation, minimum, maximum, and quartile values.

Business Interpretation

- Shipment quantity and line item value show significant variability across shipments.
- Freight-related variables exhibit wide ranges, indicating both small and high-value logistics operations.
- The summary provides an overview of data distribution and helps identify skewness, variability, and potential outliers before visualization.

Figure 20.1: Top 10 Destination Countries by Number of Shipments

Screenshot



Description

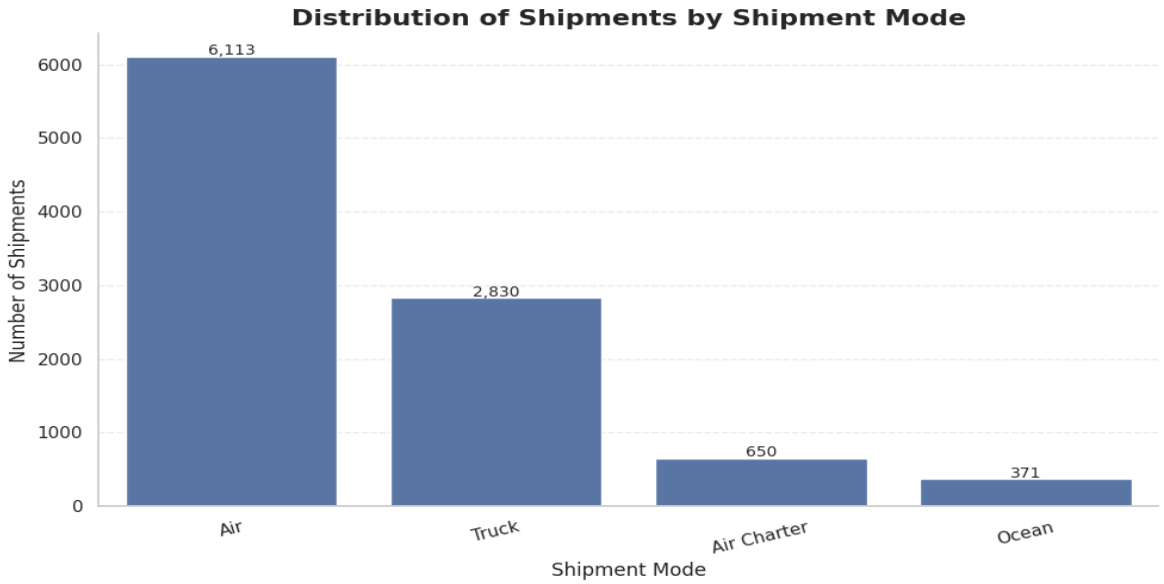
This horizontal bar chart illustrates the ten destination countries with the highest shipment volumes.

Business Interpretation

- South Africa, Nigeria, and Côte d'Ivoire account for a significant share of total shipments.
- These countries represent key logistics markets and should remain priority regions for transportation planning and resource allocation.

Figure 20.2: Shipment Mode Distribution

Screenshot



Description

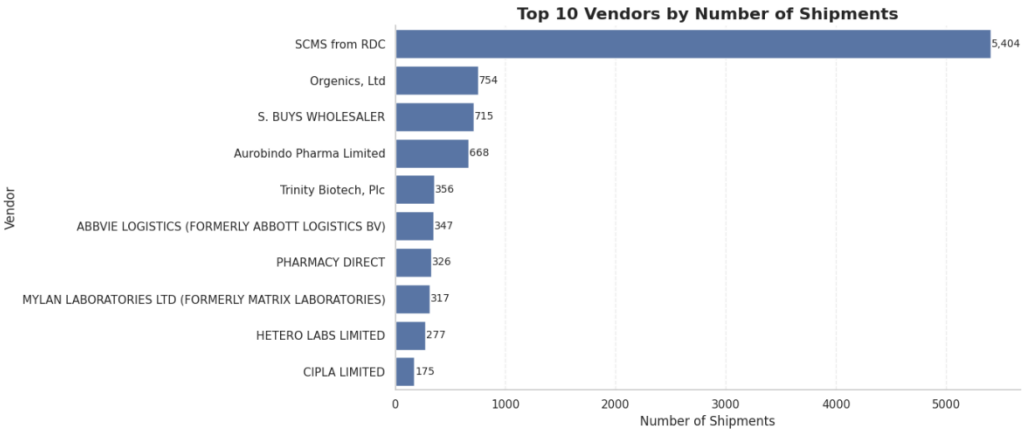
This bar chart presents the distribution of shipments across different transportation modes.

Business Interpretation

- Air transportation is the most frequently used shipment mode.
- Ocean and Truck shipments represent a comparatively smaller share, indicating opportunities to optimize transportation mode selection for cost efficiency.

Figure 20.3: Top 10 Vendors by Number of Shipments

Screenshot



Description

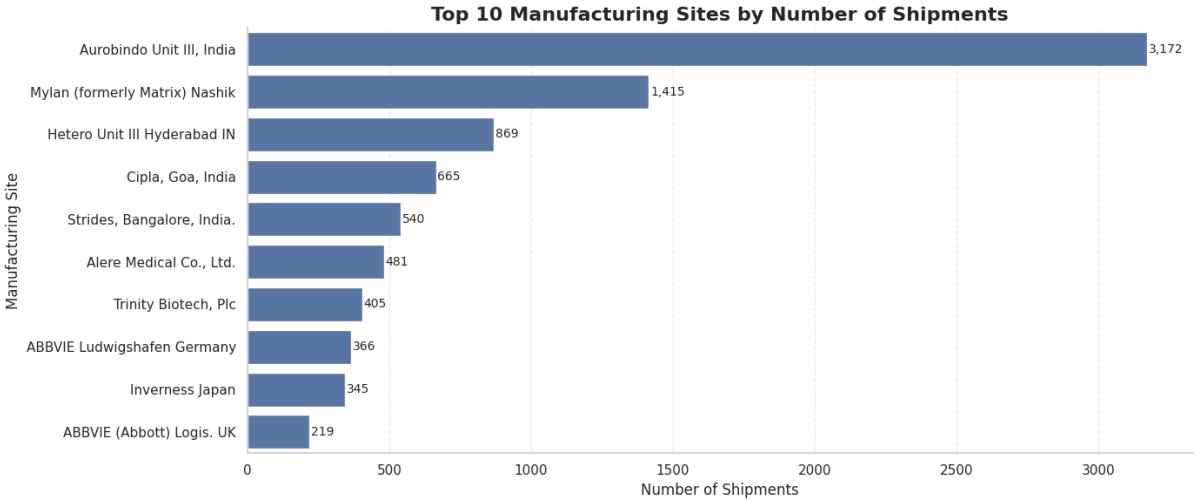
This visualization identifies the vendors responsible for the highest shipment volumes.

Business Interpretation

- A limited number of vendors contribute significantly to shipment operations.
- Strategic supplier management can improve procurement efficiency and supply chain reliability.

Figure 20.4: Top 10 Manufacturing Sites by Shipment Volume

Screenshot



Description

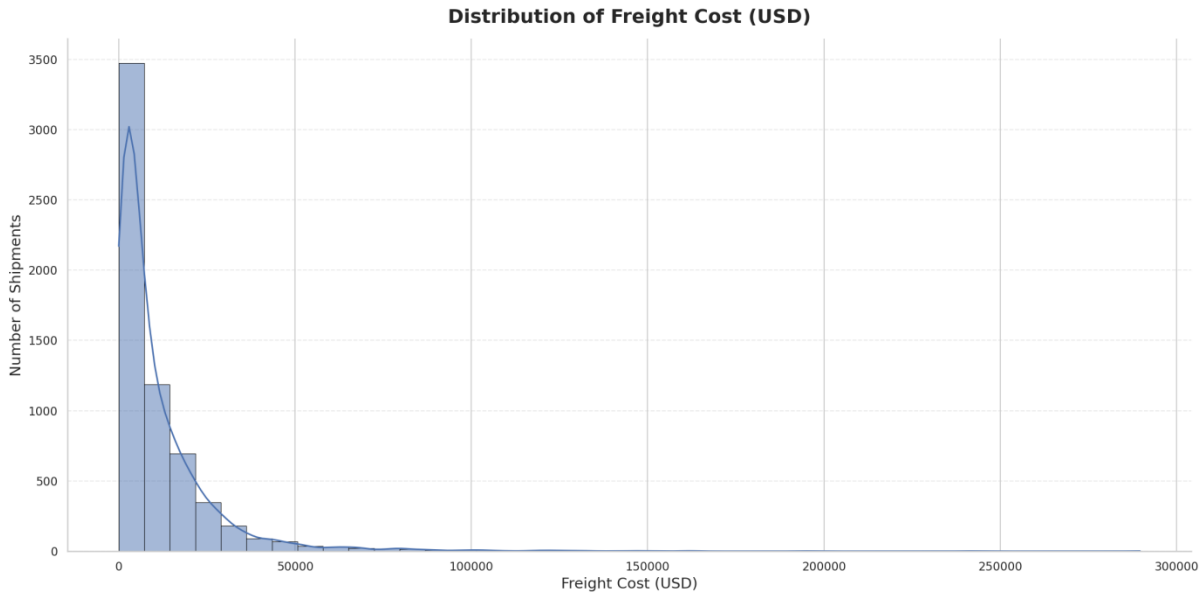
This chart highlights the manufacturing locations generating the largest number of shipments.

Business Interpretation

- Shipment activities are concentrated among a few manufacturing facilities.
- These locations play a critical role in maintaining an efficient and responsive supply chain.

Figure 20.5: Distribution of Freight Cost

Screenshot



Description

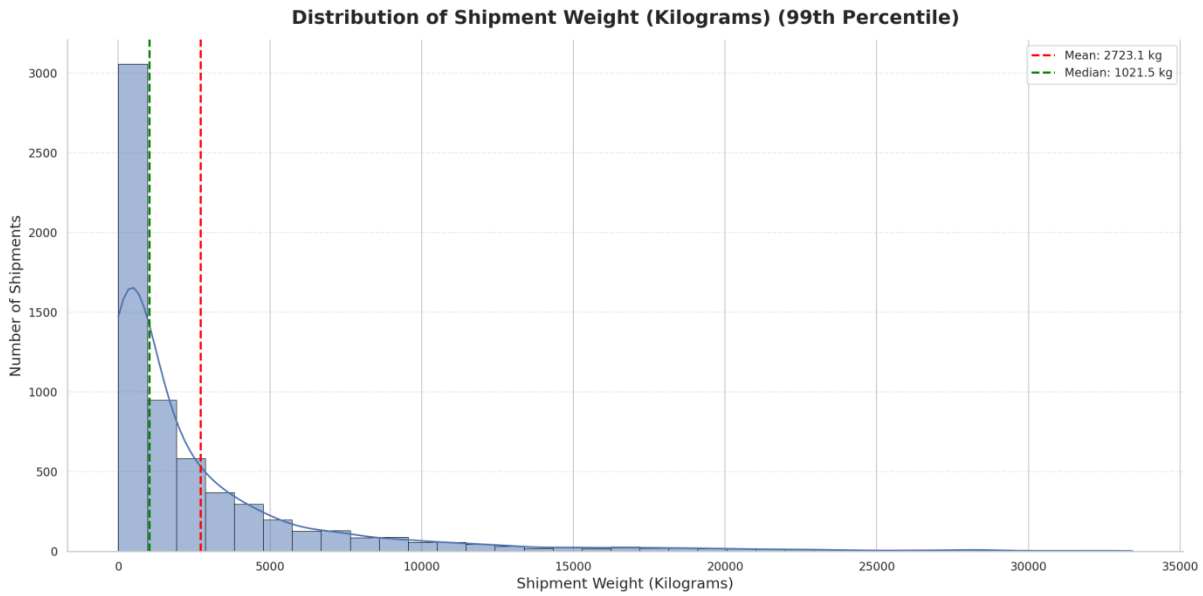
A histogram illustrating the distribution of freight costs across all shipments.

Business Interpretation

- Freight costs exhibit a positively skewed distribution.
- Most shipments incur relatively low transportation costs, while a few high-cost shipments significantly influence the overall distribution.

Figure 20.6: Distribution of Shipment Weight

Screenshot



Description

This histogram displays the distribution of shipment weights.

Business Interpretation

- The majority of shipments consist of relatively lightweight consignments.
- A small number of heavy shipments create noticeable outliers and contribute disproportionately to transportation costs.

Figure 20.7: Shipment Mode vs. Freight Cost

Screenshot



Description

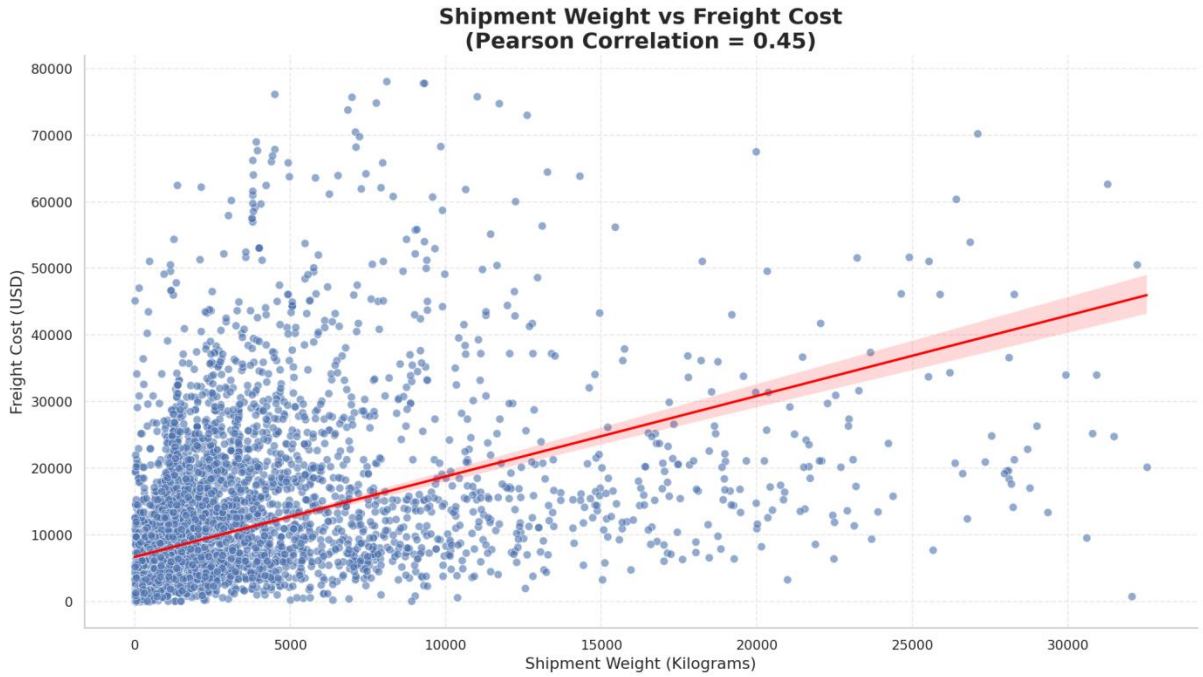
A box plot comparing freight cost distributions across different shipment modes.

Business Interpretation

- Air transportation generally incurs higher freight costs than other shipment modes.
- Selecting cost-effective transportation modes for non-urgent shipments can improve operational efficiency.

Figure 20.8: Shipment Weight vs. Freight Cost

Screenshot



Description

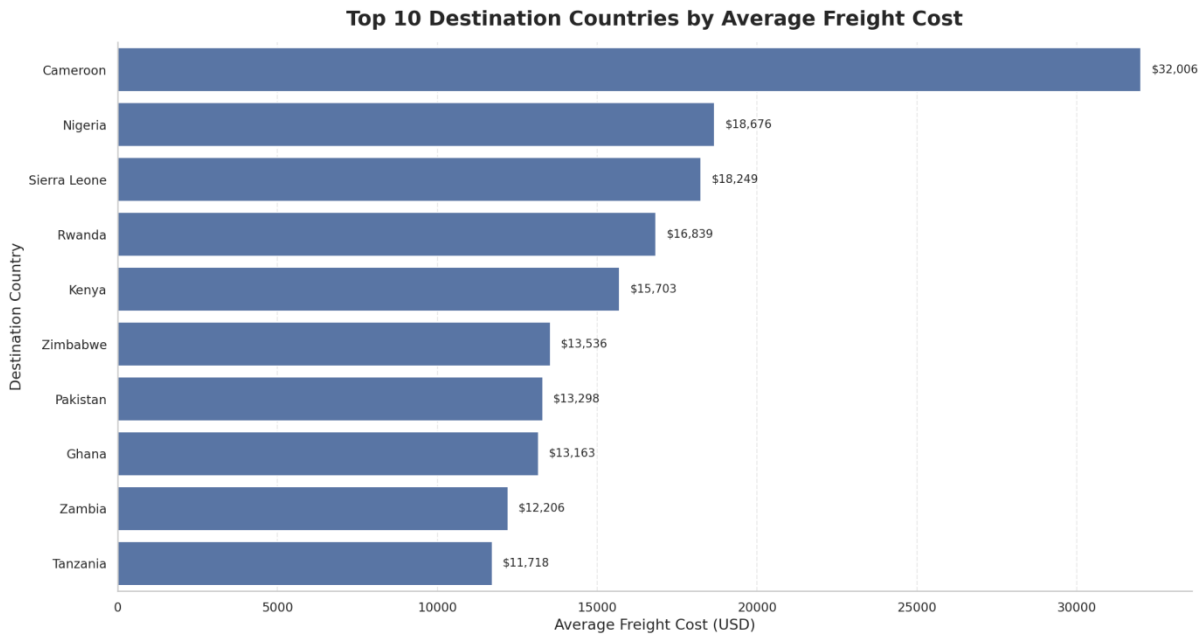
This scatter plot illustrates the relationship between shipment weight and freight cost.

Business Interpretation

- Freight cost generally increases as shipment weight increases.
- The visualization confirms a moderate positive relationship between shipment weight and transportation expenses.

Figure 20.9: Top Destination Countries by Average Freight Cost

Screenshot



Description

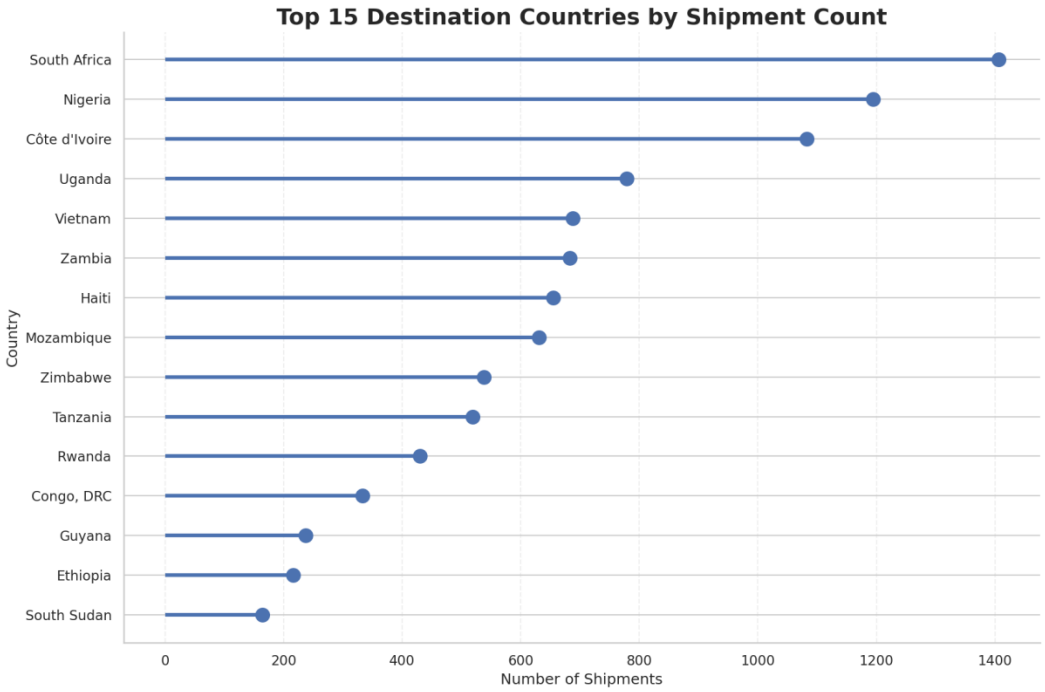
A horizontal bar chart comparing average freight costs across major destination countries.

Business Interpretation

- Certain destination countries consistently experience higher transportation costs.
- Regional logistics optimization and carrier selection can help reduce shipping expenses.

Figure 20.10: Top 15 Destination Countries (Lollipop Chart)

Screenshot



Description

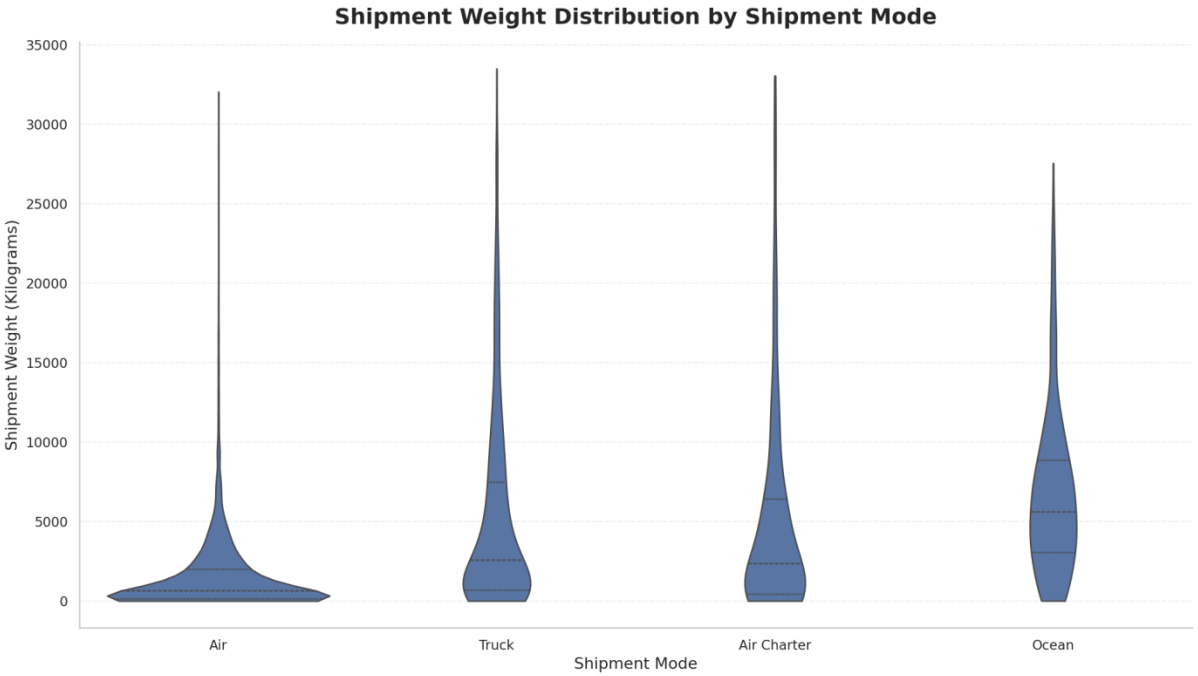
A lollipop chart presenting shipment frequency across the top destination countries.

Business Interpretation

- The visualization clearly highlights shipment concentration within a limited number of key markets.
- It provides an effective comparison of shipment volumes across countries.

Figure 20.11: Shipment Weight Distribution by Shipment Mode

Screenshot



Description

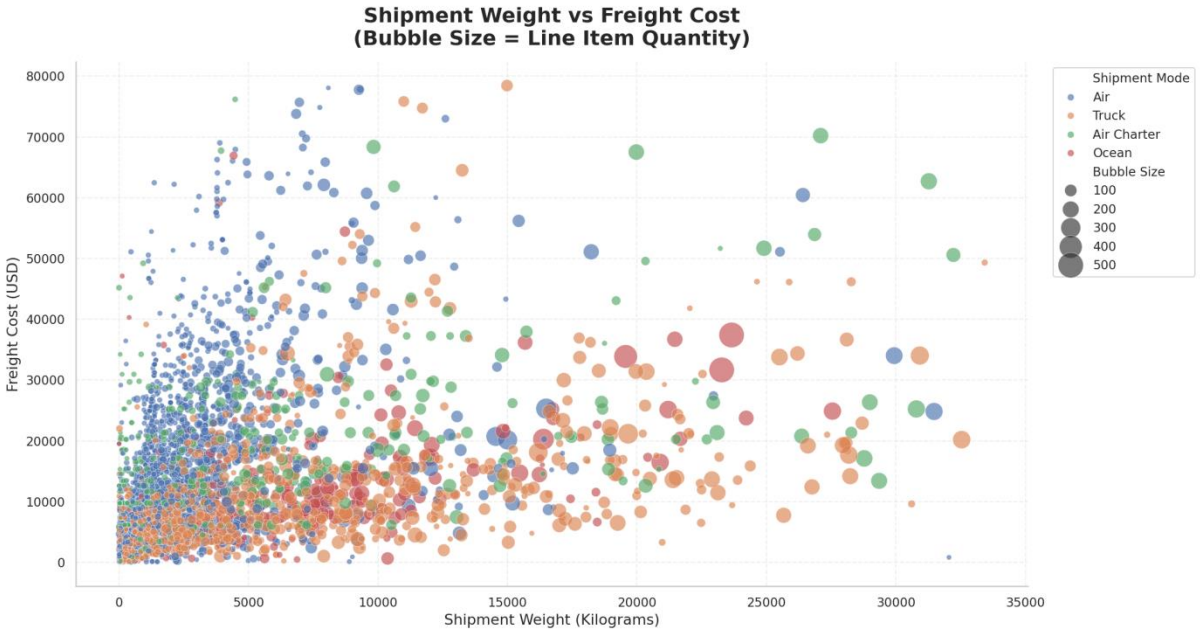
A violin plot comparing shipment weight distributions across transportation modes.

Business Interpretation

- Air shipments typically carry lighter consignments.
- Ocean transportation is more commonly used for heavier shipments, reflecting differences in transportation capacity and cost.

Figure 20.12: Shipment Weight vs. Freight Cost by Shipment Mode

Screenshot



Description

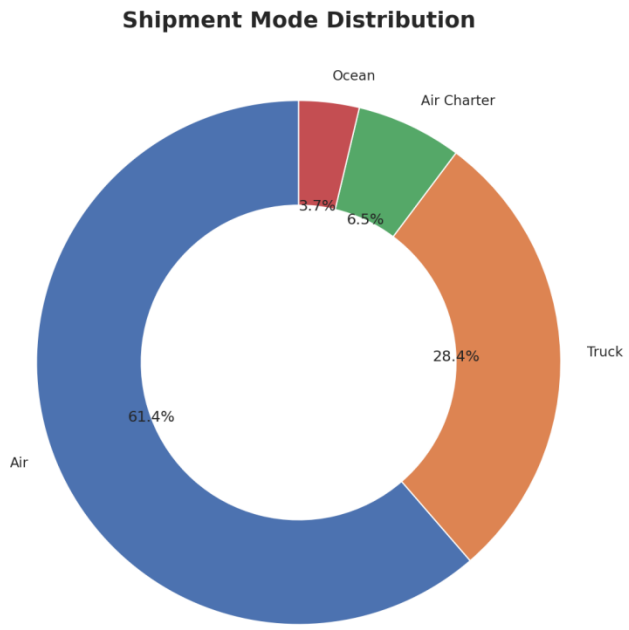
A bubble chart illustrating the combined relationship among shipment weight, freight cost, shipment mode, and shipment quantity.

Business Interpretation

- Larger shipment quantities are generally associated with higher freight costs.
- Transportation mode significantly influences the overall cost structure of shipments.

Figure 20.13: Shipment Mode Distribution (Donut Chart)

Screenshot



Description

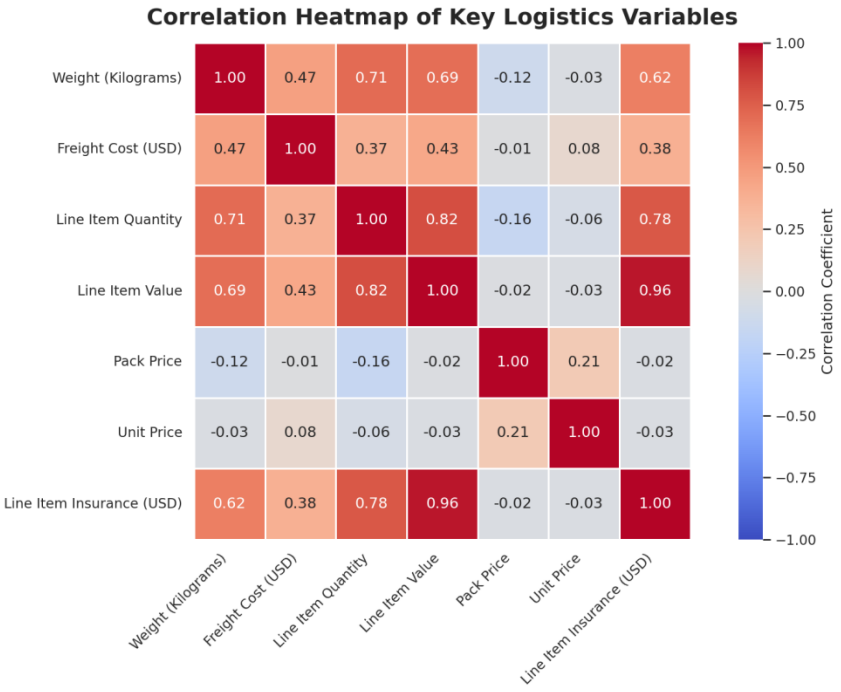
A donut chart showing the proportional distribution of shipment modes.

Business Interpretation

- Air transport represents the largest proportion of shipments.
- The visualization highlights the organization's reliance on rapid delivery services.

Figure 20.14: Correlation Heatmap of Logistics Variables

Screenshot



Description

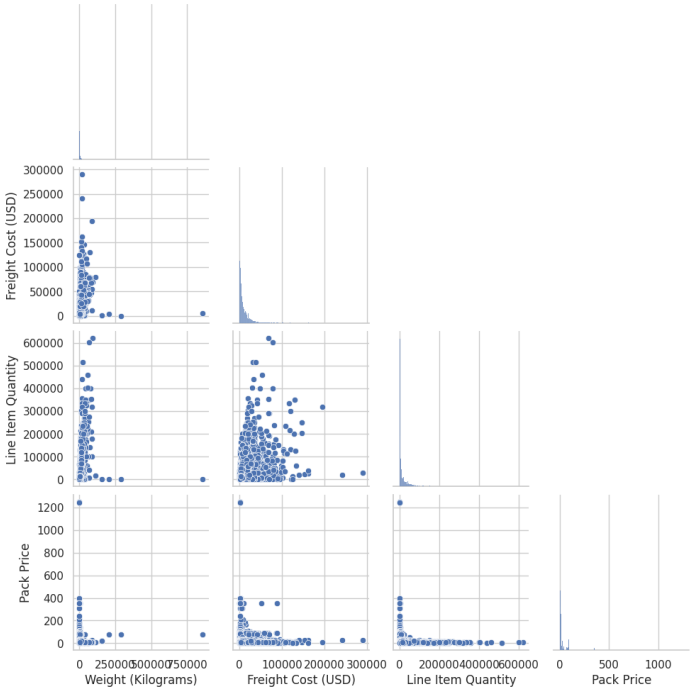
A correlation heatmap illustrating relationships among major numerical logistics variables.

Business Interpretation

- A strong positive relationship exists between **Line Item Value** and **Line Item Insurance (USD)**.
- Shipment Weight shows a moderate positive correlation with Freight Cost.
- The heatmap helps identify variables that influence transportation expenses and shipment value.

Figure 20.15: Pair Plot of Key Logistics Variables

Screenshot



Description

A pair plot displaying pairwise relationships and distributions among key numerical variables.

Business Interpretation

- The visualization confirms positive relationships among shipment weight, freight cost, shipment quantity, and line item value.
- It also assists in identifying clusters, trends, and potential outliers across the dataset.

Overall Interpretation

The collection of visualizations provides a comprehensive understanding of FedEx's logistics operations by revealing shipment distributions, transportation patterns, regional performance, freight cost behavior, and relationships among key operational variables. Collectively, these figures transform complex logistics data into actionable business insights, supporting strategic planning, cost optimization, operational efficiency, and data-driven decision-making.

These graphical analyses complement the statistical findings and form the basis for the business insights and recommendations presented in this report.

21. Conclusion

The **FedEx Logistics Operations Analysis** project successfully demonstrates the value of **Exploratory Data Analysis (EDA)** in transforming complex logistics data into meaningful business intelligence. By applying a structured analytical approach—including data understanding, preprocessing, statistical analysis, and visualization—the project provided valuable insights into shipment operations, transportation performance, freight costs, supplier contributions, and regional distribution patterns.

The analysis revealed several key findings, including the concentration of shipments in major destination countries, the predominance of air transportation, the positive relationship between shipment weight and freight cost, and the strong association between shipment value and insurance cost. It also highlighted the influence of transportation mode, shipment characteristics, and supplier networks on overall logistics performance. These insights enable a better understanding of operational behavior and identify opportunities to improve efficiency across the supply chain.

Based on the findings, practical recommendations were proposed to optimize transportation routes, improve shipment mode selection, strengthen vendor management, enhance risk mitigation, reduce freight costs, and support more effective resource allocation. The project also emphasized the importance of maintaining high-quality data and adopting data-driven decision-making practices to improve operational visibility and business performance.

Overall, this study demonstrates that Exploratory Data Analysis serves as a powerful foundation for strategic logistics planning and continuous operational improvement. The insights generated can assist decision-makers in enhancing supply chain efficiency, controlling transportation costs, improving customer satisfaction, and increasing organizational competitiveness. Furthermore, the project establishes a strong platform for future advancements through predictive analytics, machine learning, real-time monitoring dashboards, and intelligent supply chain optimization, enabling FedEx to build a more resilient, efficient, and data-driven logistics network.

22. References

The following references were consulted during the development of this project for understanding the dataset, performing exploratory data analysis, implementing Python-based analytics, and preparing the project documentation. The references are presented in **APA 7th Edition** format.

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9. Matplotlib Documentation. Matplotlib Development Team. Retrieved from the official documentation.
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Citation Style

This project follows the **American Psychological Association (APA) 7th Edition** citation style for books, technical documentation, online resources, and software references. All websites referenced are official sources, and all books cited are widely recognized references in the fields of data analytics, logistics, supply chain management, and data science.

23. Appendix

The appendix presents supplementary information that supports the analysis, methodology, and findings of the **FedEx Logistics Operations Analysis** project. It provides additional technical details regarding the dataset, software environment, preprocessing procedures, analytical techniques, visualization summary, project deliverables, and repository information. These materials enhance the transparency, reproducibility, and overall completeness of the project while serving as a valuable reference for future research and advanced analytics.

Appendix A: Dataset Information

Attribute	Details
Dataset Name	SCMS Delivery History Dataset
Domain	Logistics & Supply Chain Analytics
Source	Kaggle (SCMS Delivery History Dataset)
File Format	CSV
Total Records	10,324
Total Features	33
Numerical Features	7
Categorical Features	26
Duplicate Records	0
Total Missing Values	2,383
Memory Usage	16.04 MB

Purpose:
The dataset contains historical shipment records related to supply chain and logistics operations, including shipment details, destination countries, transportation modes, vendors, manufacturing sites, freight costs, insurance values, delivery schedules, and product information. It serves as the primary data source for performing Exploratory Data Analysis (EDA).

Appendix B: Software and Development Environment

Component	Specification
Programming Language	Python
Development Platform	Google Colab
Operating System	Windows 11
Data Processing	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn, Plotly
Notebook Format	Jupyter Notebook (.ipynb)
Version Control	GitHub

Purpose:
The project was developed using a cloud-based Python environment, enabling efficient data preprocessing, statistical analysis, visualization, and collaborative development.

Appendix C: Python Libraries Used

The following Python libraries were utilized throughout the project:

- **Pandas** – Data loading, cleaning, transformation, and analysis.
- **NumPy** – Numerical computation and array operations.
- **Matplotlib** – Static data visualization.
- **Seaborn** – Statistical data visualization.
- **Plotly** – Interactive visualizations.
- **SciPy** – Statistical functions and numerical analysis.
- **Warnings** – Suppression of unnecessary runtime warnings during execution.

Appendix D: Data Preprocessing Activities

To ensure data quality and analytical accuracy, the following preprocessing tasks were performed before conducting Exploratory Data Analysis:

- Imported the dataset into the Google Colab environment.
- Created a working copy of the original dataset.
- Standardized column names by removing unnecessary whitespace.
- Cleaned text-based fields to eliminate formatting inconsistencies.
- Converted delivery-related date columns into the appropriate datetime format.
- Identified and analyzed missing values across all features.
- Converted text-based numerical fields into numeric data types where applicable.
- Verified the absence of duplicate records.
- Reset the dataframe index after preprocessing.
- Prepared the cleaned dataset for statistical analysis and visualization.

Appendix E: Exploratory Data Analysis Summary

The analysis followed the **Univariate–Bivariate–Multivariate (UBM)** framework.

Univariate Analysis

- Shipment distribution by destination country
- Shipment mode distribution
- Vendor distribution
- Manufacturing site distribution
- Freight cost distribution
- Shipment weight distribution

Bivariate Analysis

- Shipment Mode vs. Freight Cost
- Shipment Weight vs. Freight Cost
- Average Freight Cost by Destination Country
- Shipment Weight Distribution by Shipment Mode

Multivariate Analysis

- Bubble Chart
- Correlation Heatmap
- Pair Plot

Appendix F: Summary of Visualizations

A total of **17 analytical visualizations** were developed during the project.

Figure	Visualization
20.1	Missing Values by Feature
20.2	Statistical Summary of Numerical Features
20.3	Top 10 Destination Countries by Number of Shipments
20.4	Shipment Mode Distribution
20.5	Top 10 Vendors by Number of Shipments
20.6	Top 10 Manufacturing Sites by Shipment Volume
20.7	Distribution of Freight Cost
20.8	Distribution of Shipment Weight
20.9	Shipment Mode vs. Freight Cost
20.10	Shipment Weight vs. Freight Cost
20.11	Average Freight Cost by Destination Country
20.12	Top 15 Destination Countries (Lollipop Chart)
20.13	Shipment Weight Distribution by Shipment Mode (Violin Plot)
20.14	Shipment Weight vs. Freight Cost by Shipment Mode (Bubble Chart)
20.15	Shipment Mode Distribution (Donut Chart)
20.16	Correlation Heatmap of Logistics Variables
20.17	Pair Plot of Key Logistics Variables

Appendix G: Key Findings

The Exploratory Data Analysis produced several important business insights:

- Air transportation is the most frequently used shipment mode across the logistics network.
- Shipment weight exhibits a moderate positive relationship with freight cost.
- High-value shipments require proportionally higher insurance coverage.
- South Africa, Nigeria, and Côte d'Ivoire represent the highest shipment-volume destinations.
- No duplicate records were identified, indicating good data consistency.
- Missing values were limited to a small number of variables and were appropriately handled during preprocessing.
- Freight cost and shipment weight exhibit positively skewed distributions with several high-value outliers.
- A relatively small number of vendors and manufacturing sites account for a significant proportion of total shipment activity.

Appendix H: Project Deliverables

The final project submission includes the following components:

- Exploratory Data Analysis Notebook (.ipynb)
- Final Project Report (PDF/Word)
- SCMS Delivery History Dataset (CSV)
- Python Source Code
- Graph Visualizations
- Business Insights and Recommendations
- Microsoft PowerPoint Presentation (PPT)
- GitHub Repository

Appendix I: GitHub Repository

Project Title:
FedEx Logistics Operations Analysis

GitHub Profile:
https://github.com/Ayushdash02

Repository Contents:

- Complete EDA Notebook
- Dataset
- Python Source Code
- Visualizations
- Project Report
- Presentation Slides
- Project Documentation

Appendix Summary

The appendix provides comprehensive technical documentation supporting the **FedEx Logistics Operations Analysis** project. It includes detailed information on the dataset, development environment, preprocessing methodology, analytical workflow, visualizations, project deliverables, and repository resources. These supplementary materials enhance the project's transparency, reproducibility, and technical credibility while providing a solid foundation for future enhancements, including predictive analytics, machine learning applications, real-time dashboard development, and advanced supply chain optimization.